

# Yilin LIU

Curriculum Vitae

Peking University, Beijing, China

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## Summary

- Third-year Physics undergraduate at Peking University, working on high-energy physics data analysis within the **LHCb** experiment.
- Experience in heavy-flavor physics analysis including **amplitude analysis** and  **$CP$  violation measurements**.
- Hands-on experience with **ROOT**, **Python**, and the **LHCb analysis framework** (e.g., DaVinci).

## Education

Sep. 2023 – Now

**Bachelor in Physics**, School of Physics, Peking University

Expected Graduation: 2027

- Rank: Top 15% of the physics cohort
- Selected Coursework
  - *Particle Physics* (A+)
  - *General Relativity* (95)
  - *Mathematical Methods in Physics (1)* (Full marks)
  - *Precision Measurement in the Frontiers of Physics* (A)
- Honors & Awards
  - Merit Student, Peking University, 2024
  - Merit Student, Peking University, 2025

## Relevant Experience

**Undergraduate Research (Advisor: Prof. Yanxi Zhang)**

December 2023 – Present

- Contributed to the amplitude analysis of the  $B^+ \rightarrow \psi(2S)K^+\pi^0$  decay.
  - Developed the full analysis workflow including event selection, signal extraction, and amplitude analysis.
  - Extracted signal yields using unbinned extended maximum likelihood fit to the invariant mass of  $\psi(2S)K^+\pi^0$ .
  - Calculated phase-space distributions and evaluated the efficiencies.
  - Performed amplitude analysis using a TensorFlow-based fitting framework.
- Search for  $CP$  violation in  $B^0 \rightarrow K^+a_1^- (\rightarrow \pi^+\pi^-\pi^-)$  decay.
  - Built workflow from event selection to signal extraction; extracted signal yields using unbinned extended maximum likelihood fit to the invariant mass of  $K^+\pi^+\pi^-\pi^-$ .
  - Measured the direct  $CP$  violation and Triple Product Asymmetry (TPA).
  - Determined the production asymmetry of  $B^0$  using the control channel  $B^0 \rightarrow \pi^+D^-(K^+\pi^-\pi^-)$ .
  - Evaluated experimental asymmetries including detection, trigger, and particle identification asymmetries.
  - Presented analysis updates in the **BnoC** working group of **LHCb** collaboration.
  - Supported by the undergraduate research program of **National Natural Science Foundation of China (NSFC)**.

## Technical Skills

|                       |                                                               |
|-----------------------|---------------------------------------------------------------|
| <b>Analysis:</b>      | ROOT (histograms, cutflow, efficiency studies, RooFit), NumPy |
| <b>Programming:</b>   | Python                                                        |
| <b>LHCb Software:</b> | DaVinci                                                       |
| <b>Workflow:</b>      | Linux                                                         |

## Languages

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**English:** Fluent in academic reading and presentations (CET-6);   **Chinese:** Native